

MCCOOK, NE, USA

WMO: 725625

Lat: 40.206N

Lon: 100.591W

Elev: 2530

StdP: 13.40

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 94040

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.5	5.3	-7.2	4.1	2.4	-2.1	5.4	8.1	33.3	41.9	28.8	41.7	8.3	250	0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.2	100.0	70.6	97.0	70.4	93.2	70.0	75.3	89.6	73.8	88.9	72.4	87.5	13.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.6	128.4	82.9	69.8	120.6	80.9	67.9	112.7	78.7	40.6	89.4	39.1	88.8	37.7	87.5	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.9	25.7	22.3	-8.3	105.6	4.9	3.5	-11.8	108.1	-14.7	110.1	-17.4	112.1	-21.0	114.6		
(5)			-8.9	77.6	4.8	1.2	-12.3	78.5	-15.1	79.2	-17.7	79.9	-21.2	80.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.9	29.6	31.6	42.2	51.1	61.5	72.7	78.5	75.5	67.5	52.8	40.3	29.6		
	DBStd	19.73	10.97	11.32	10.56	9.53	8.67	7.09	5.90	5.65	8.49	9.58	9.76	10.54		
	HDD50	2571	634	519	282	98	12	0	0	0	2	79	311	634		
	HDD65	5654	1099	935	706	423	172	15	1	3	71	390	741	1098		
	CDD50	3613	1	4	41	132	367	681	884	789	527	167	20	0		
	CDD65	1222	0	0	0	7	63	246	420	327	146	13	0	0		
	CDH74	15124	0	4	69	227	976	3003	5085	3563	1887	292	18	0		
	CDH80	7542	0	0	13	61	383	1478	2821	1810	886	89	1	0		
(14) Wind	WSAvg	10.1	9.6	10.3	11.2	12.5	11.1	10.4	9.0	8.4	9.5	10.1	9.7	9.5		
(15) Precipitation	PrecAvg	20.5	0.5	0.6	1.3	1.9	3.0	3.2	3.3	2.6	1.6	1.2	0.9	0.5		
	PrecMax	29.5	1.5	2.6	4.2	5.8	6.9	7.8	7.9	6.1	7.1	4.9	3.7	2.0		
	PrecMin	9.7	0.0	0.0	0.0	0.1	0.0	0.4	0.7	0.1	0.1	0.0	0.0	0.0		
	PrecStd	4.1	0.4	0.6	1.1	1.4	1.8	1.8	1.8	1.5	1.5	1.1	0.8	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.3	71.4	81.3	87.2	94.9	102.0	105.5	102.0	99.2	89.5	76.8	63.3		
		MCWB	45.8	49.7	56.3	57.9	64.2	69.9	70.3	69.1	67.6	62.8	53.9	46.2		
	2%	DB	57.9	62.6	74.5	80.6	89.0	97.7	100.6	97.4	93.5	81.6	69.9	55.7		
		MCWB	43.1	45.8	52.3	57.7	64.4	69.9	71.4	70.0	67.7	60.1	51.5	42.4		
	5%	DB	52.0	55.2	68.2	74.9	83.9	93.0	97.5	93.3	90.1	76.0	63.4	51.0		
		MCWB	40.3	42.6	50.3	55.9	63.2	69.4	72.0	70.8	66.7	57.6	48.5	40.2		
	10%	DB	46.0	49.6	62.5	69.7	79.4	88.5	93.4	90.3	84.6	70.7	57.2	45.5		
		MCWB	37.1	40.1	47.8	53.9	61.5	68.4	71.6	70.6	64.7	55.6	45.0	36.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.7	50.5	59.9	62.4	70.1	75.5	77.4	76.8	73.0	66.5	56.5	48.3		
		MCDB	64.3	65.7	71.2	78.7	84.8	88.8	92.9	90.9	88.4	79.7	69.1	59.5		
	2%	WB	43.5	46.6	55.2	59.7	67.3	73.0	75.9	74.7	70.3	62.7	52.7	43.5		
		MCDB	57.2	60.2	68.4	74.7	82.3	89.9	90.6	88.7	86.3	75.1	65.2	55.5		
	5%	WB	40.7	43.6	51.5	57.1	65.3	71.3	74.5	73.1	68.6	59.8	49.5	40.4		
		MCDB	51.4	54.4	66.4	71.3	79.6	87.5	89.5	87.3	83.9	72.3	62.5	50.4		
	10%	WB	37.4	40.3	48.3	54.9	63.1	69.7	73.0	71.6	67.1	57.0	46.0	37.2		
		MCDB	45.2	49.4	61.6	68.2	75.7	85.4	88.5	85.3	81.2	68.3	57.4	45.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	24.2	23.7	27.9	26.5	25.9	26.2	26.2	25.5	28.1	27.2	27.0	23.5		
		MCDBR	34.2	34.8	39.9	36.9	33.4	32.5	31.8	31.0	33.8	36.9	37.0	32.6		
	5% WB	MCWBR	21.2	21.0	21.0	17.9	14.0	11.5	10.4	10.9	12.4	17.5	20.7	20.7		
		MCWBR	32.6	33.8	35.7	31.2	28.3	28.3	27.2	26.4	28.7	29.8	35.2	31.3		
(40) Clear-Sky Solar Irradiance	taub		0.263	0.270	0.297	0.328	0.357	0.367	0.390	0.378	0.346	0.305	0.271	0.261		
	taud		2.542	2.531	2.471	2.408	2.401	2.407	2.384	2.405	2.449	2.538	2.554	2.539		
	Ebn at Noon		293	305	305	300	292	288	280	281	284	287	287	285		
	Edh at Noon		23	26	31	36	37	37	37	36	32	26	22	21		
(44) All-Sky Solar Radiation	RadAvg		754	1026	1408	1733	2001	2280	2246	1953	1588	1149	833	657		
	RadStd		58	80	115	94	131	119	85	104	84	115	54	43		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page